

Crabs on camera: code documentation

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Background

Project and repository aims

The basis of this project is rooted in the work done in the masters thesis by Brøten (2026). The aim of the project was to improve monitoring of the data-limited Brown crab (*Cancer pagurus*) stock in Norway using machine learning and underwater video surveys.

The aim of this repository is not to push the boundaries of machine learning or showcase advanced coding, rather to make these methods more approachable for marine scientists working in fisheries applications.

The brown crab supports a valuable small-scale inshore fishery of ecological, economic and social importance to coastal communities. Despite this, data limitations persist and constrain stock assessments. Traditional methods for collecting fisheries independent data are manual, and prone to bias (hiddink_estimating_2024?). The Institute of Marine Research (IMR) already runs coastal monitoring programs and surveys that collect underwater video footage (BRUV) across many sites along the coast. This footage is a large, currently underutilized resource for stock monitoring, but a bottleneck persists due to the manual annotation process (Durden *et al.* 2016). This project explores whether a machine learning model can automate that review step, using video data collected through IMR's existing coastal monitoring programs and surveys to train a crab detector.

About the repository

This repository uses the same code, functions, and packages developed for the masters thesis, though some functions have been simplified here to make the pipeline easier to reproduce.

To try it yourself, clone this repository — either on your own computer for the small example dataset, or on a server with GPU access if you plan to train on a larger dataset.

Annotations in this project were created in the Computer Vision Annotation Tool (CVAT; Seachev et al, 2020) using RLE (run-length encoded) masks.

AI tools (large language models) were used throughout this repository to help generate code, fix syntax, and clean up errors.

Before running the scripts

Prerequisites

Python must be installed. `raw_to_yolo.ipynb` checks for missing packages and installs them automatically (see Chunk 0).

A server with GPU access is recommended if you plan to train on a large dataset — the small example dataset shipped with this repo can be run locally without one.

If you want to train a detector on your own data, you'll need an annotated dataset structured the same way as the example dataset: an `annotations` folder containing the CVAT/COCO export (`instances_default.json`), and an `images/default` folder containing the matching image files.

Data

The repo only ships a small example/test dataset (`data/annotations/test_annotations`) so the scripts can be run and verified without downloading everything.

The full training dataset (~8,000 images) used in the original masters project is hosted externally: **[LINK TO ADD]**. Download it and point `CONFIG["ANNOTATIONS_FILE"]` / `CONFIG["IMAGES_DIR"]` in `raw_to_yolo.ipynb` at the local copy to reproduce the full training run.

From raw annotations to YOLO-ready data

The aim of the first script is to lead you from raw annotations to a dataset ready for YOLO

On the stratified sampling in the k-folds - if your data is balanced, and you're not worried about station leakage - then this part can be dropped.

Chunks 0 - 2

Downloads all the necessary packages to run the

Brøten G. Crabs on camera: Improving the monitoring of the brown crab (*Cancer pagurus*) with machine learning and Baited Remote Underwater Video (BRUV). Master's thesis, University of Bergen, 2026. <https://hdl.handle.net/11250/5537346>.

Durden J, Bett B, Schoening T *et al.* Comparison of image annotation data generated by multiple investigators for benthic ecology. *Marine Ecology Progress Series* 2016;**552**:61–70. <https://doi.org/10.3354/meps11775>.

Sekachev B, Manovich N, Zhiltsov M *et al.* *Opencv/Cvat: V1.1.0*. 2020. <https://zenodo.org/records/4009388> (7 Aug. 2026, date last accessed).